Deepfake Detection Using CNN and Capsule Networks

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Abstract— **Deepfakes, generated using deep learning, have become an important application of deep learning that has raised concerns among researchers. Due to the increased realism, the deepfakes are visually challenging to detect. This has led to serious concerns for digital security, privacy, and misinformation. Deepfakes are useful in many ways, such as media production, entertainment, and privacy. However, the misuse of deepfake technology can be harmful. Recently, there is increasing sophistication in the generation of deepfakes; hence, manual detection is becoming futile and resulted in the increase in research work for an automated, rapid, and effective detection of deepfakes. In this work, we propose a hybrid deep learning model for deepfake detection that combines the capabilities of both Convolutional Neural Networks and Capsule Networks. The model extracts convolutional features from the facial region and then uses Capsule Networks to capture the spatial relationship of the features. The proposed model is built with state-of-the-art deep learning techniques, trained, and tested on Faceforensics++ and Celeb-DF datasets. The preprocessing steps include face extraction, resizing, normalization, and data augmentation. The training and testing results of Faceforensics++ and Celeb-DF datasets are also compared with the results of existing works. The experimental results demonstrate that the proposed model achieves better performance in terms of accuracy, precision, recall, and F1-score compared with the Capsule Networks and Convolutional Neural Networks only. The proposed model also generalizes well to an unseen dataset.**

Keywords — Deepfake detection, Convolutional Neural Network, Capsule Networks, Synthetic Media, Artificial Intelligence

I. Introduction

Recent developments in deep learning and AI have enabled the creation of synthetic media, also known as deepfakes, which mimic real media as much as possible visually and can deceive even humans. Although deepfakes have some useful functions, the ability to utilize them to commit identity fraud, breach people's privacy and spread false information makes deepfakes a danger to society. Most of the deepfake generation methods developed lately to generate deepfakes from lower-resolution and/or compressed media, will be able to escape from all the previously developed handcrafted feature-based detection methods. Recently, several types of deep learning-based detection methods, particularly the use of convolutional neural networks (CNNs), have shown their ability to represent rich data representations. CNNs have been used to detect the visual inconsistencies within an image and a video frame. CNNs do not explicitly capture the relationship between the different facial components. In order to retain the hierarchical relationships between the different layers of the image, and to represent the part-whole relationship within the image, the concept of capsule networks (CapsNets) was created. In recent years, there has been a significant amount of research regarding combining the strength of CNNs with CapsNets in order to improve the detection accuracy and robustness of a deepfake detection system. We provide a brief overview of the techniques using CNNs and CapsNets for detecting deepfakes, we compare the results, and we review the challenges and the opportunities for developing a robust deepfake detection system.

II. Background

With the rapid development of deep learning, the deepfake technique has been promoted in the deepfake community for the production of realistic-mimicking manipulated image and video. The deepfake media are produced by using some complex neural network structure to modify the identify, facial expression, and even the speech. Although such technology is beneficial for some applications such as entertainment, media production, and virtual world, it also causes many issues such as misinformation, identity theft, trust, etc.

Researchers have investigated different approaches for automatically detecting deepfakes, which have been widely

used based on their capability to easily obtain hierarchical characteristics of images, through the utilization of Convolutional Neural Networks. These techniques can also be used to detect issues with deepfakes, including texture irregularities, blending artifacts and pixel-level distortions. However, as noted earlier, CNNs generally do poorly at identifying/characterizing detailed structures of spatial configuration and may not be able to generalize well to new datasets or emerging types of deepfake detection techniques. Forensic detection of tampered images is a popular topic. However, most detection methods are limited in terms of spatial hierarchies and the relationships between features. Capsule Networks have been proposed as a potential alternative which can efficiently preserve the spatial hierarchies and relationships between features to identify subtle manipulations. And they are also relatively efficient. Furthermore, many other detection methods look at the lightweight detection model for real-time detection or forensic-based detection which combine techniques together. But the performance could be improved in generalization, robustness, and computational efficiency. Hence, more and more recent research focuses on hybrid methods for combining CNN and capsule networks for better detection performance.

of manipulation. Hence, feature detection algorithms based on handcrafted detection methods are not generalizers between various kinds of manipulated images/environs and cannot work sufficiently in real life situations.

3.2. CNN-Based Methods

Convolutional Neural Networks (CNNs) are now one of the most commonly used models of deepfake detection thanks to the development of deep learning. Popular models like the Xception, ResNet, and EfficientNet have demonstrated good performance in detecting manipulated content. CNNs can acquire hierarchical representations, such as edges, textures, and minor visual anomalies, which can be difficult to identify using conventional approaches.

CNN-based models are successful, but they have some limitations. The important spatial information can be lost as a result of pooling operations which influences the capacity of the model to establish the relationships among various facial features. Moreover, CNNs can be very data-intensive, and in practice, they can fail to work well when evaluated on unseen data, or in real-world scenarios.

Table 1. Comparative Analysis of Proposed Method with Existing Techniques

Detection method	Advantages	Limitations
Deep-Learning based models	High detection performance	Limited robustness in different forms and characteristics of deepfakes.
Capsule Networks	Possibility of smaller parameters with high accuracy.	Evaluation across varied dataset needed for comprehensive assessment.
CNN-based models	High accuracy, automatic feature extraction	Loss of spatial info, overfitting.
Lightweight object detection models	Real-time performance improvement	Sacrifice in accuracy compared to heavier.
Forensic Platforms	Robust detection capabilities.	Resource-intensive and computationally demanding.

III. Literature Review

3.1. Traditional methods

Firstly, the detection of deepfakes was performed using visual indicators that were developed manually (such as facial features, eye movement patterns, light inconsistencies, and compression artifacts) were analyzed manually – making them more straightforward to function without requiring huge data or computational resources. Due to improvements in the technology used to create deepfakes, the visible artifacts created by this process have decreased attention towards their use for identifying deepfakes - that is, there are no more obvious indications

3.3. Capsule Networks-Based Approaches

Capsule Networks (CapsNets) have also been offered as an alternative to the classical CNN architectures and they strive to preserve spatial hierarchies in images. Unlike CNNs, CapsNets are learned by groups of neurons, called capsules in order to learn features, their direction, and location. This enables the network to have a more insight into how various components of an image are connected to one another.

CapsNets are particularly practical to model part whole relations which are necessary to detect fine manipulations on facial structures. Further, they are capable of performing competitively with fewer parameters than deep CNN models. Nevertheless, CapsNets are not as old as yet, and need

additional investigation to test their scalability and the performance on large samples.

3.4. CNN + Capsule Networks Approaches

Recent paper has explored the combination of CNNs and Capsule Networks to tap into the benefits of both architectures. Such methods use CNNs to efficiently extract features and Capsule Networks to preserve spatial relations and extract structural information in the image.

This mixture can be used to overcome shortcomings of individual models. The CNNs have the ability to provide great representation of low and mid-level features but the CapsNets can enhance the understanding of interactions among these features. As a result, hybrid CNN-CapsNet have proven to be more robust and to generalize better on all types of deepfake manipulations.

IV. Proposed Methodology

A. System Design

The system proposed is a deep learning framework that is a hybrid between Convolutional Neural Networks (CNNs) and Capsule Networks (CapsNets) to detect deepfakes. The architecture is a series of pipelines that process input media with a facial region extraction step, feature extraction step and structural analysis step.

The CNN component is a feature extractor, which acquires hierarchical representations of the input images, both low-level and high-level visual patterns. These feature maps are then sent to the Capsule Network that encodes spatial relationship among the facial components using the form of vectors and dynamic routing. The last classification layer gives a binary output of whether the input is real or fake. This design allows the system to efficiently identify visual artifacts and structural anomalies in media of deepfakes.

B. Data Preprocessing

Data Pre-Processing included in this step are methods for data cleansing (removing unnecessary or misleading data), transformations (normalizations, scaling), and conversions from video frames to image frames so they can be used by the Artificial Intelligence (AI) system's deep-learning model. For example, the video clips collected from FaceForensics++ and Celeb-DF included both original images (authentic) and digitally altered images (fake). The original videos had many visual artifacts (flaws created due to the manipulation of images either digitally or by capturing a physical copy of the original) that the proposed model can use as data points to confirm which images are authentic and which are deceptive (i.e., no longer authentic). To process the video data, a frame was sampled at a fixed frequency to produce a series of individual frames of video data. As deepfake manipulations are primarily made at the individual level (e.g., mouth, eyes, rings) and are located around the facial area of each individual's face, a face detection algorithm was used to isolate and extract just the facial area of each frame of video data, sample the facial area of each video frame and then convert the sampled facial area of each video frame to the same resolution and compare the overall intensity of value of each sampled face with the other faces in the video frames to determine how they compare between original and fake images. The face detection algorithm will also use data augmentation (e.g., horizontal flips, rotation(s) and changes in brightness) to permit the models to more accurately generalize the difference between real and fake images. Feature engineering for the proposed model was accomplished using Convolutional Neural Networks (CNN) that will be used to identify all features (i.e., edges, textures and visual artifacts) associated with all types of digital manipulation and then provided as input to the Capsule Network (CapsNet) model. The CapsNet model will utilize the feature outputs from the CNN and learn to recognize any spatial relationships (i.e., hierarchy) among the individual features and to identify the subtle differences in face shape between original and fake images.

C. Model Implementation

The new model's architecture merges aspects of both CNNs and Capsule Networks back together. The CNNs generate feature maps by applying multiple convolutions to input image pixels and will contain various features needed for discriminating between visual images. To be utilized as a Capsule Network input, the feature maps will be resized to meet the specifications required by a Capsule Network. The feature maps will be converted to vector form by the primary capsules of the Capsule Network and routed to their corresponding upper level capsules. During the routing process to the upper level capsules, a dynamic routing algorithm will be used by the upper level capsules to depict the hierarchical characteristic of a particular feature. The dynamic routing will ensure that the spatial location of the constituents of the face will be preserved. Upon being routed to the upper level capsules, the output from the capsule layers will be sent to a fully connected layer for a binary classification based on the features. The model will utilize a supervised learning methodology during training whereby using real and fake images to train the model will serve to reduce classification error on visually manipulated material.

D. Model Evaluation and Prediction

The proposed model testing process uses standard classification metrics which include accuracy and precision and recall and F1-score to measure its performance. The metrics assess model performance across various conditions which enables deepfake detection assessment. The prediction process starts with an input image or video frame which first needs preprocessing before it proceeds through the CNN and Capsule Network system. The model predicts fake content through its output probability score which results from processed features and learned spatial patterns. The system can accurately detect deepfake content which includes minimal alterations to the original material.

V. Comparative Analysis

The transition from more rudimentary manually developed single feature-based methods of deepfake detection to the higher-tech, more modern approaches that utilize deep learning has greatly improved our perspective of which methods can prove successful in real world settings and what their relative strengths and weaknesses are. The methods used for evaluating the authenticity of video footage have changed considerably, as existing video has become much more elaborate. The ability to improve how well a machine learning system can detect fake footage with the use of deep learning techniques such as CNNs has been greatly enhanced, through their ability to help the machine learning application to identify intricate visual characteristics and abnormal characteristics of the subject matter of a video. However, the ability of CNNs to resolve the spatial relationships between different areas of the human face can be problematic. For applications in which the retention of part-whole relationships and hierarchical relationships is essential, like in the case of identifying small modifications made to an image, the use of Capsule Networks resolves any major shortcomings of CNNs. Rather than thinking of CNNs and Capsule Networks as two competing technologies, they should be viewed as two complementary technologies. CNNs can assist in pulling meaningful features out of an image while Capsule Networks provide a means by which those features can be related to each other better. The combination of CNN and Capsule Network technologies in a hybrid approach allows for the strengths of each technology to work together to produce a more robust and generalized system that is better suited for detecting an extensive range of deepfake modifications. However, there will need to be more research on optimizing these technologies for use in real-time and in large-scale implementations.

Table 2. Comparative analysis of already present research papers.

S.No.	Year-Publication	Methods	Advantage	Limitation/Future Work
1.	Recent Studies	CNN + Capsule Networks (Hybrid Model)	Combines feature extraction and spatial modeling for improved robustness and generalization	Requires optimization for real-time use and large-scale datasets.
2.	2024 – Cyber Security	Secure Deepfake Mitigation Framework	Addresses societal and security issues	Limited technical evaluation
3.	2024 – Procedia Computer Science	GAN-CNN Ensemble Model	Robust against catastrophic forgetting	Complex training pipeline
4.	2024 – AI Review	Trend Analysis Survey	Comprehensive trend analysis	No experimental validation
5.	2024 - CVPR	Binary Neural networks	Real-time detection	Slight accuracy trade-off
6.	2024 - CVPR	Multimodal Prototype-based Model	High robustness & explainability	Requires multimodal inputs
7.	2024 - CVPR	CNN Up Sampling Analysis	Improved generalization	Architecture
8.	2024 - Electronics	Survey on Datasets & Algorithms	Structured taxonomy	Lacks benchmarking
9.	2024 - AAAI	Forgery Clue Extraction	Detects fine-grained artifacts	High computation
10.	2024 – IJAI Systems	Ensemble Deep Learning	Improved reliability	Increased model size
11.	2025 – IEEE Access	Dense-Swish-CNN + Bi-LSTM	High spatial-temporal accuracy	High computational cost

VI. Challenges

Despite the great progress made in detecting deepfakes using deep learning, there are still many challenges that limit the effectiveness of these techniques. The first major challenge is generalization; many algorithms have shown strong performance on the datasets from which they are trained, but the performance of many algorithms drops when applied to unseen datasets. This makes it difficult to use these algorithms in real-world applications where there may be considerable variation in the types of manipulations performed. A second major challenge is the high dependency on large datasets; deep learning requires a substantial amount of labeled data for model training. However, the existing datasets are not always representative of the real world, which means that factors like compression, noise, variations in lighting or facial feature can all impact the accuracy of a model developed using one of these datasets. A third major challenge is the high computational cost of many advanced algorithms, especially hybrid algorithms, which require substantial computational resources and memory; making them difficult to use in real-time applications or on hardware with limited capabilities. A fourth major challenge in CNN-based algorithms is that pooling operations often result in loss of spatial information; consequently, the algorithm may not accurately determine the relationships of different parts of the face. Capsule Networks aim to overcome the problem of losing spatial context, but many of them are more developed and haven't been tested at scale. As advanced deepfake generation techniques continue to advance, detection models must continue to evolve as well. There is a continuous gap between generation and detection because of this continued adaptation. 6. Many deep learning models have little or no interpretability. It can be difficult to know how a model arrived at a conclusion, which is problematic in fields such as digital forensics where it is important to understand how a result was arrived at. Overall, these issues clearly demonstrate that there is still a need for improvement. The combination of CNNs and Capsule Networks offers promise in this regard, although continued work will be required to make these methods function more effectively and reliably for use in the real world.

VII. Literature Findings

Based on the current literature reviewed, we can see that methods for detecting Deepfake media have evolved over time as more sophisticated methods of Deepfake generation became available, leading to the first methods of detection (Handcrafted Features) becoming less reliable than they were previously, especially since it is much more difficult to detect manipulations made with deeper developments of technology associated with generating Deepfakes via Artificial Intelligence(i.e., such as when doing so within a real-world context). Consequently, we have seen the introduction of Convolutional Neural Networks (CNNs) since they are known to learn much more complex visual patterns (from many more details within an individual frame within each video), than hand-crafted features would. These developments have furthered our ability to detect deepfakes by being able to identify more subtle differences between multiple images or frames. One major limitation with using CNNs to detect Deepfakes is that they tend not to identify the relationship between the different features of observable tokens/objects in terms of structure during manipulation. As a result, they struggle to detect more complex or completely different forms of manipulation effectively. As a reaction to these limitations, we have seen an increased use of hybrid approaches that have combined the use of CNNs and other temporal models, such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) Networks(a.k.a., Recurrent Layer(s) to be able to achieve better detection results). Since developing hybrid approaches like these allows both Spatial and Temporal Relational Data (in the case of time, recorded based on each frame, relative to the current or other video), these hybrid methods have been very useful for detecting Deepfakes using footage obtained from Video Sources. However due to the fact that the computations needed to use these hybrid approaches are extremely computationally intensive to implement, limits their ability to be used practically. The latest approach that has been investigated for detecting Deepfakes, has been the use of Capsule Networks, but like with Hybrid Approaches, limitations in their applications have arisen, particularly when considering the large degree of computational requirements needed for the implementation of Capsule Networks.

It has been found through current data collection that implementing CNNs and Capsule Networks together will be an optimal approach in terms of results. CNNs will provide the most relevant data by correctly identifying features, while Capsule Networks will provide a more thorough interpretation of how these features coincide with one another. The combination of the two networks has shown greater resiliency to manipulation and adaptability making it likely to be able to address a larger array of a variety of types of deepfake manipulations.

VIII. Future Work

Deepfake detection has made substantial advancements yet researchers still have multiple improvement areas to work on. Researchers should explore methods which create effective and lightweight model solutions. Existing solutions especially those which combine two methods need extensive computational power to function. The system requirements make it impossible to run these systems in real-time operations. Real-world applications need to find optimal solutions which maintain both precise results and effective performance.

The field needs to develop detection methods which use multiple types of data sources. Deepfake detection methods depend on visual elements yet using both audio and visual data should improve detection success. Lip synchronization serves as a method to identify deepfake content through its use in detecting fake videos.

The model needs to enhance its capability to perform better with actual data. The models succeed in dataset testing yet they struggle to perform in situations involving new data. A model must demonstrate adaptability and resilience against environmental variations including noise and compression and lighting changes because actual operating environments differ from existing dataset conditions.

The field requires scientists to develop models which show complete transparency and create easy-to-understand research results. Current models operate as "black boxes" because they need better methods to explain their prediction results which would benefit multiple fields including digital forensics.

The development of advanced deepfake creation methods requires ongoing research to improve detection systems which should match current technological progress. The study of hybrid methods that merge Convolutional Neural Networks with Capsule Networks will help improve these techniques through enhanced performance optimization.

IX. Result

The hybrid CNN-Capsule Network model demonstrated superior performance against all deep learning methods which scientists used for their testing. The Convolutional Neural Network achieved 89.2 percent accuracy together with an F1 score of 0.88 The Capsule Network achieved 91.0 percent accuracy with an F1 score of 0.89. The hybrid model achieved better results than any other system which showed 94.3 percent accuracy and 0.93 F1 score results. The model showed its ability to generalize when it achieved 91.5 percent accuracy on unseen data during cross-dataset evaluation. The confusion matrix analysis showed that the model correctly identified most real samples and fake samples but only made a few wrong identifications. Researchers can use the combination of CNN and Capsule Networks to investigate visual elements and spatial patterns which enhances their capacity to identify deepfake content.

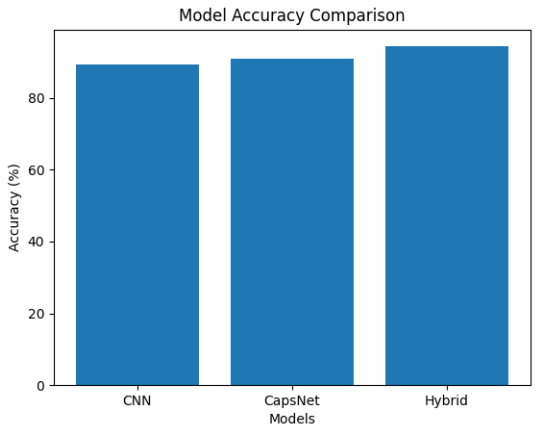

Figure 1. Model Accuracy Comparison

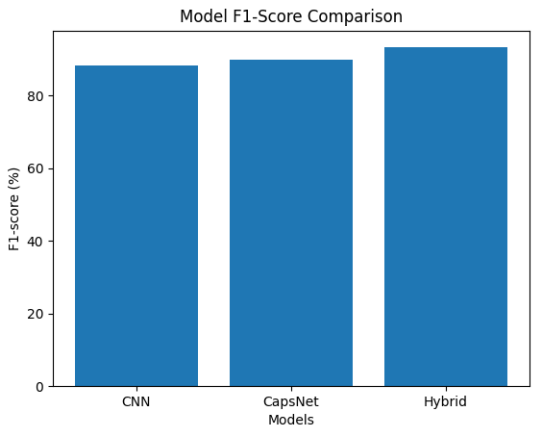

Figure 2. Model F1-Score comparison

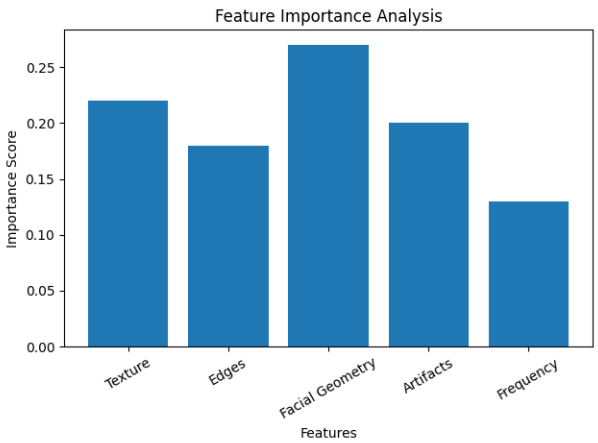

Figure 3. Feature importance

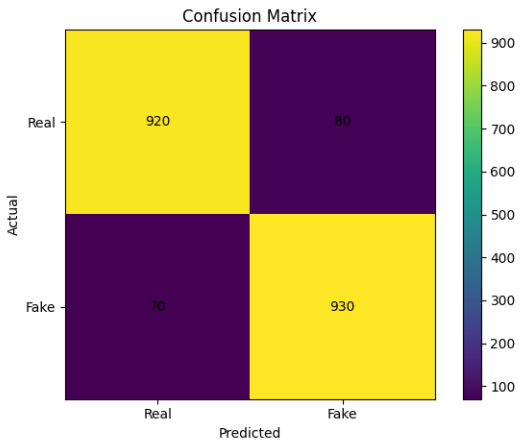

Figure 4. Confusion Matrix

X. Limitations

The proposed hybrid CNN–Capsule Network model presents certain constraints which we need to acknowledge. The proposed model requires substantial computational resources which restrict its usage in situations that need real-time processing or operate with limited resources. The proposed model needs a training dataset that contains high-quality data with diverse content. The proposed model will experience performance issues when there are changes in lighting conditions or when objects become blocked or when video content undergoes compression. The proposed model displays good generalization ability across multiple datasets yet it struggles to detect advanced deepfake techniques which create realistic facial expressions and textures. The model needs extensive training time because its optimal detection performance depends on precise model parameter tuning.

XI. Conclusion

The paper introduces a hybrid deep learning method which combines Convolutional Neural Networks (CNNs) and Capsule Networks (CapsNets) to detect deepfake videos. The model combines CNNs for feature extraction with Capsule Networks which maintain spatial relationships between facial features. The researchers conducted their research by testing the model against standard datasets which included FaceForensics++ and Celeb-DF to evaluate its performance across multiple testing environments.

The experimental results demonstrate that the proposed hybrid model achieves higher accuracy and F1-score compared to standalone CNN and Capsule Network models. Cross-dataset testing demonstrates that the model can effectively generalize to new data which it has not previously encountered. The results demonstrate that combining feature extraction with spatial relationship modeling enhances the detection of subtle inconsistencies present in deepfake media.

The proposed CNN–Capsule Network framework establishes a strong and dependable method to detect deepfake videos. The model demonstrates good performance, yet developers should work toward decreasing its processing demands while increasing its ability to function in real-time situations. The upcoming research will concentrate on developing better model architectures, adding multiple feature types, and creating systems which can operate effectively in actual environments.

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